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INSTRUMENTATION AND CONTROL IN SAFETY APPLICATIONS

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1.0 SCOPE

US and international standards organizations have recently issued standards for evaluating and maintaining Safety Integrity Levels (SIL) for Electrical / Electronic / Programmable Electronic Systems (E/E/PES) for safety applications. These systems are intended to keep process systems within safe operating boundaries. It supplements detailed design guidance found in documents such as IEC 61508 or ANSI/ISA S84.01.

This data sheet outlines an approach to the specification, design, installation, operation, maintenance and modification of Safety Instrumented Systems (SIS). It provides a methodology to specify the SIL needed based on incident frequency and consequences. The recommendations have been organized to approximate the safety system life cycle of Hazard Assessment, Management Systems, Hardware, Software, Acceptance Testing, Operation and Maintenance. Basic criteria to be considered with respect to location construction and protection of control or equipment rooms is also provided. It does not provide comprehensive checklists nor state in detail how to design, operate or maintain these systems. This information should be obtained from appropriate ANSI/ISA and IEC sources noted in Section 4.0.

Although not intended as design criteria for the Basic Process Control System (BPCS), many of the steps listed in this data sheet are equally applicable in design of such systems.

E/E/PES are found in almost every industrial occupancy including automotive, power and gas utilities, cogeneration plants, petrochemicals, pulp and paper, metals, mining, food processing, packaging, lumber, coal, machine shops, etc.

1.1 Definitions

Actuator — See Output Device

Analog — Signals that vary continuously over a range (e.g., temperature, pressure) rather than one discrete or digital signal that switches between high and low values (e.g., limit switches).

BPCS (Basic Process Control System) — A system that responds to input signals from the equipment under control (EUC) and/or from an operator and generates output signals, causing the EUC to operate in the desired manner. Examples include control of an exothermic reaction, anti surge control of a compressor and fuel/air controls in fired heaters. Also referred to as Process Control System.

Certified — Process by which an SIS, or SIS component, is examined and tested by a third party (FMRC or other accredited agency) for compliance with applicable standards (e.g., ISA S84.01, IEC 61508) to determine Safety Integrity Level (SIL).

DCS (Distributed Control System) — A network of programmable electronic systems performing monitoring, controlling, safeguarding and archival (database) functions, via common communication links, for the one or more processes being managed. As an electronic process management system, it does not require the redundancy and reliability of a safety system. Recent trends include integrating the safety function tightly with the DCS, which then requires additional reliability be provided for the DCS. (see 2.3.1, recommendation 2)

Demand rate — The frequency with which a protective system is called upon to perform its protective function.

Diagnostics — A troubleshooting program for identifying hardware or software malfunctions.

Diversity — Different means of performing a required function, for example, other physical principles, other ways of solving the same problem.

E/E/PES — Electrical / Electronic / Programmable Electronic Systems — A system based on one or more E/E/PE devices, connected to (and including) input devices (e.g., sensors) and/or output devices/final elements (e.g., actuators), for the purpose of control, protection or monitoring. Figure 1 shows a simple PES and typical components.

Note: *The term is intended to cover any and all devices operating on electrical principles and would include:*
electro-mechanical devices (“electrical”);
solid state non-programmable electronic devices (“electronic”);
electronic devices based on computer technology (“programmable electronic”)

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Emergency Stop — Ability to shut down a process independently of safety and process control systems.

Fail-safe — Capability to go to a predetermined safe state in the event of a specific malfunction.

Field devices — Equipment connected to the field side of the SIS I/O terminals. Such equipment includes field wiring, sensors, final control elements and operator interface devices hard-wired to SIS I/O terminals.

Final Control Element — A device that manipulates a process variable to achieve control.

Hardwired Systems — Systems are not programmable and require physical changes to wiring or links to change their function. Examples include relay logic and non-programmed electronic logic as well as fiber optic communications links.

Independent Protection Layer (IPL) — A mechanical system or subsystem specifically designed to reduce the likelihood or severity of an identified hazardous event by a large factor, i.e., at least by a 100 fold reduction in likelihood. (See also 3.1.2)

Initiator — A device or combination of devices that indicates whether a process or equipment item is operating outside the operating envelope, i.e., a set point is reached. The Initiator includes input cards and input relays. Examples are manual switches, position switches and measurement systems (including process connections, sensors, transmitters, cabling, trip amplifiers or input cards, etc.).

Input Device — An external device that reads process or control variable values (i.e., thermocouple, level switch, etc.) and provides a signal for the PES, usually through an input module.

Input Module — A device that converts information received from the input device to digital information understandable to the PES. It may be an analog-to-digital converter or, in the case of simple "yes" or "no" inputs (from limit switches, etc.) it may just alter the voltage to that used in the logic of the PES.

I/O (Input or Output) — This all-encompassing term describes input and output modules and devices, or a particular mode of input or output.

Logic Solver — The E/E/PES components or subsystems that execute the application logic. The Logic Solver excludes trip amplifiers, input cards and output cards. Examples are electromechanical relays, solid-state/magnetic-core logic and the Central Processing Unit (CPU) section of programmable electronic systems.

Output Device — An external device that takes a signal created as a result of a PES decision and performs an action (i.e., fuel shutoff valve, steam turbine governor valves, etc.).

Output Module — A device that either converts digital output from the PES to analog output, or changes the digital output voltage to a level sufficient to operate the output device.

PES — (Programmable Electronic Systems) — A system based on one or more central processing units connected to sensors and/or actuators on a plant for the purpose of control, monitoring or protection.

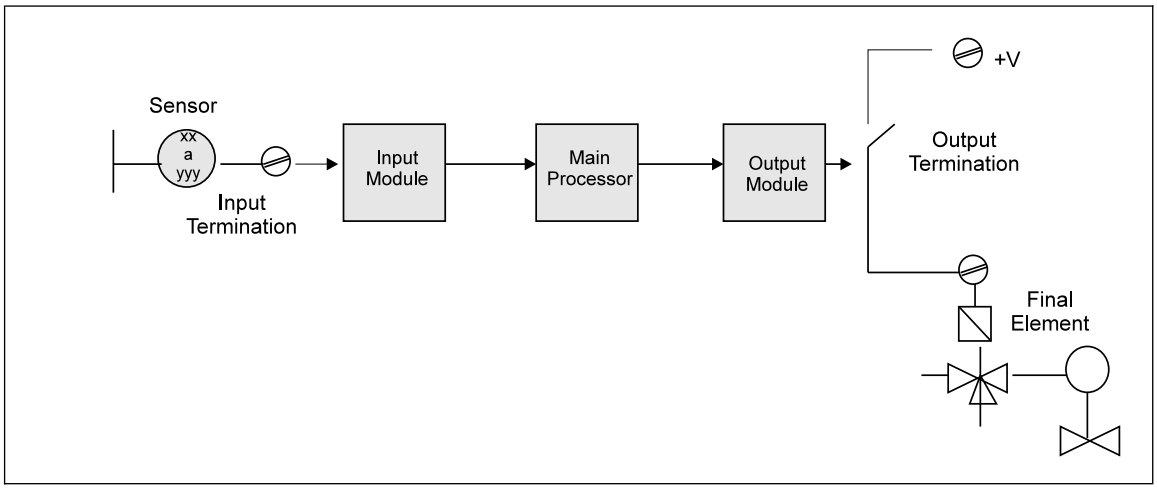


Fig. 1. Single Programmable Electronic (PE) system with single I/O, single sensor and final element.

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PLC (Programmable Logic Controller) — A component of a PES. A digitally operating electronic apparatus using programmable memory. Memory stores instructions and implements specific functions such as logic, sequencing, timing, counting, and arithmetic to control various types of machines or processes via digital and analog input/output devices.

Process Control System — See BPCS

Redundancy — Use of multiple elements or systems to perform the same function. Redundancy can be implemented by identical elements (identical redundancy) or by diverse elements (diverse redundancy).

Reliability — Probability that a system can perform a defined function under stated conditions for a given period of time.

Risk — The product of the consequence of an accident times the likelihood (probability) of that accident occurring. (see also 3.1.3)

Safety Integrity Level (SIL) — One of four possible discrete levels for specifying the safety integrity (reliability) requirements of the safety functions in safety-related systems. SIL 4 has the highest level of safety integrity and SIL 1 the lowest.

Safety System — See Safety Related System.

Safety Related System — Protect property or equipment. In addition to SIS, this could include fire suppression systems, over pressure protection equipment, dikes, etc.

Sensor — See input device

Separation — The use of multiple devices or systems to segregate control from safety functions. Separation can be implemented by identical elements (identical separation) or by diverse elements (diverse separation).

SIS (Safety Instrumented Systems) — Composed of sensors, logic solvers, and final control elements designed to take the process to a safe state when predetermined conditions are violated. Other commonly used terms include Emergency Shutdown System (ESD, ESS), Safety Shutdown System (SSD), Safety System, and Safety Interlock System.

Survivability — Any requirement for a SIS to remain functional during or after a major incident.

Voting System — Redundant system which requires at least m of the n channels to be in agreement before the SIS can take an action (e.g., *m* out of *n*, one out of two [1oo2] to trip, two out of three [2oo3], etc.).

2.0 RECOMMENDATIONS

2.1 Hazard Assessment

1. Prior to writing a specification for and installing a SIS, conduct a study that evaluates the overall process hazards, determines current levels of protection and determines where improvements are needed to reach a desired level of safety. This analysis can be done by any of the following methods:

   a. Standard — a process may have well defined hazards and therefore a defined set of safety controls. An example is the Recommended Good Practice: Safe Firing of Auxiliary Fuel in Black Liquor Recovery Boilers (BLRB) with safety controls specified by the BLRB Advisory Committee.

   b. HAZOP — Hazard and Operability Study or similar methodology is conducted to develop a qualitative determination of hazards involved. The study would highlight safety concerns and possible safety system needs on a qualitative basis.

   c. QRA — Quantitative Risk Assessment is similar to HAZOP except the safety concerns are quantified using accepted or calculated incident probability and reliability data on the instruments or other mitigating methods.

2. Compare the results of the hazard assessment study with planned or available protection levels. Determine whether a predefined (desired) level of risk has been achieved by any combination of layers shown in Figure 2.

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3. To minimize the likelihood of a single failure in a ***process control system*** creating a demand on the SIS, identify all reasonably foreseeable failure modes to enable them to be addressed in the design of the ***process control system***.

For example, the I/O cards of the BPCS generally accommodate multiple channels. As they are the interface to the field devices, they are more subject to external influence (EMI, RFI, leakage currents, etc.) and subsequently fail at a greater rate than logic or processor cards. I/O must be functionally assigned such that signals to or from redundant process trains or channels do not share the same card. This will improve the overall reliability of the BPCS system and reduce the demand on the SIS.

4. Where a SIS is needed to achieve the desired overall level of risk, use the SIL Matrix, Figure 3, to determine the SIL to be specified. If the Matrix has no SIL recommended, the BPCS and other non-SIS protection layers are to be relied on for operation safety, however, at a minimum, a manual emergency stop system independent (hardwired) of any programmable controller should be provided.

5. Where the needed SIL exceeds level 4, the risk reduction needs should be shifted back to other non-SIS safety layers shown in Figure 2.

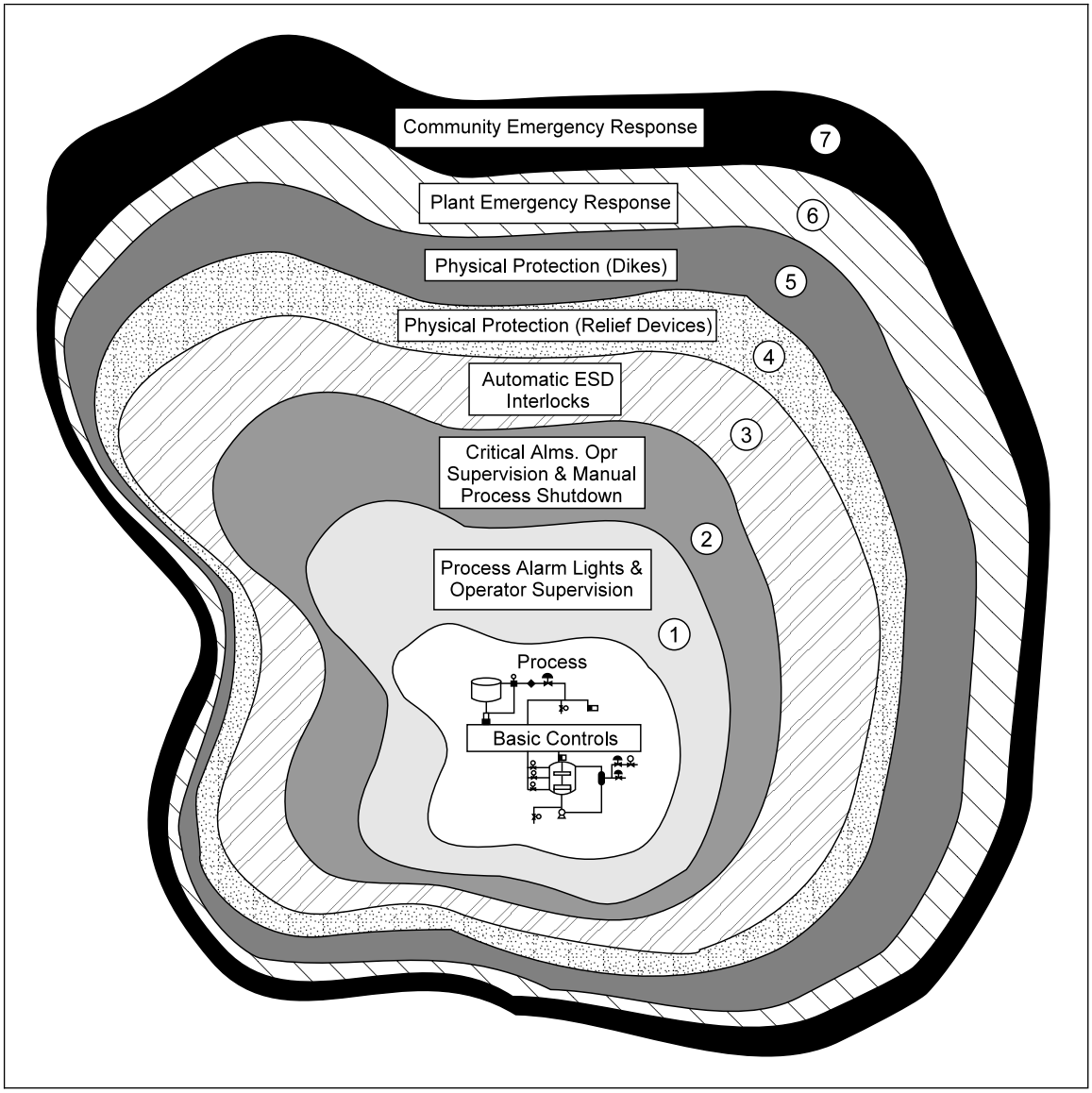


Fig. 2. Typical layers of protection in modern chemical plants.[1]

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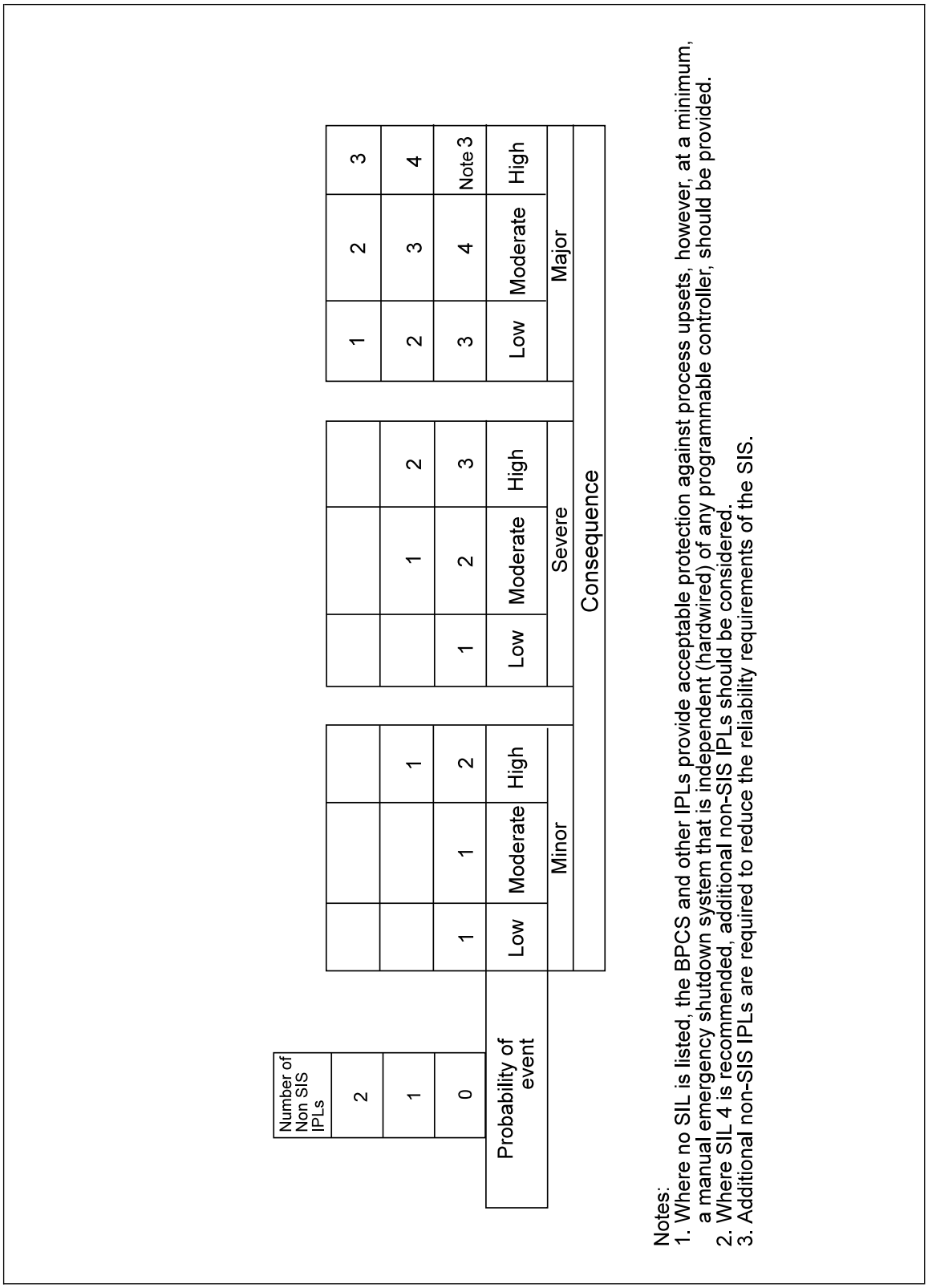


Fig. 3. SIL matrix.

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6. The SIL for the installed or proposed SIS should be determined by expert evaluation from a qualified consultant such as Factory Mutual Research Corporation (FMRC), or licensed Instrumentation and Control (I&C) Engineers.

2.2 Management Systems and Personnel Competence

Management systems and people play a crucial role in the specification, design, auditing, operation, maintenance and testing, and modification of the SIS and in ensuring that the systems meet their required performance standards. The existence of a good Management of Change program at the plant will go a long way toward ensuring the initial and continued reliability of these systems. (See also 2.6.2)

1. Have management systems and procedures in place, with a single point of ultimate responsibility to ensure that all activities associated with the SIS are planned, executed, and documented to maintain the required performance.

2. Ensure that personnel involved with the SIS know and understand the nature and extent of their responsibilities and are competent to perform them.

   a. Ensure that competence criteria includes a statement of all skills required and provides a rating for each skill.

   b. Re-assess competence annually and when the individual's responsibilities or capabilities change.

   c. Provide training, as needed, to meet the minimum competency rating required.

2.3 Hardware

2.3.1 Design and Selection for Safety

1. Base SIS design on recognized standards such as the Instrument Society of America (ISA), the International Electrotechnical Committee (IEC) or other equivalent national standard and this data sheet.

2. Separate the SIS from the Basic Process Control Systems. If combined into a common system (such as a DCS), the BPCS must meet the same design, documentation, access limitations, SIL, etc. as a designated SIS. Steps must also be taken so that failures or problems experienced by the BPCS do not compromise the SIS.

3. The failure to safety principle is preferred for all SIS components. In order to achieve this, consider each item of plant equipment to ensure predictability of failure modes. However, for certain applications, an energized to trip (non fail-safe) design concept may be justified. If so, additional measures must be taken to ensure the safety integrity of these devices, e.g., line monitoring, built-in fault detection, and/or redundancy.

4. Hardwire operator actuated emergency stop functions and keep them independent of the SIS and BPCS, however, such emergency stop commands should be communicated to the SIS and BPCS. This will prevent the systems from trying to maintain the application in operating mode and reduce the potential for a hazardous restart condition.

5. Select and design systems to minimize complexity and meet the required performance standards. Increased complexity could reduce reliability and the level of understanding by operators. Higher inspection, test and maintenance requirements may also result from complex system design.

6. Field devices tend to have a lower reliability than logic systems, therefore, specifying logic systems with a higher than needed SIL may lead to a complex system with little or no overall improvement in system SIL.

7. Specify each element of the system to performance standards consistent with the overall required functional, safety integrity, and survivability performance standards, and not simply to the highest level achievable.

8. The scan and diagnostics cycle time of the SIS should not exceed the critical time for the process. For example, a turbine could go out of control in seconds and a SIS with scan and diagnostics cycle times on the order of 1 minute could fail to detect the incident and shut down the unit safely.

9. Use only certified systems for SIL 3 and 4 applications.

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10. For SIL 3 and 4 applications, resetting the equipment under control (EUC) should only be possible at the EUC. For SIL 1 and 2 applications, equipment can be reset from a central control room. For example, a tripped shutoff valve can be returned to service from the control panel for a SIL 1 or 2 application. In a SIL 3 or 4 application the operator would go out to the valve for resetting. The intent is to require examining the equipment to discover a reason for the trip before a critical, high reliability safety system is reset.

11. Have administrative procedures and other measures in place to ensure that only appropriate authorized personnel have access to change control data or programs. If access is controlled by password, change the password at appropriate intervals under the control of the designated responsible person.

2.3.2 Environmental Considerations

1. SIS equipment selection should account for anticipated environmental conditions which could affect sensitive electronic equipment such as dirty air, temperature excursions, excessive vibrations, electromagnetic interference, etc.

2. Include fire, blast and dropped object protection in the safety system specification where appropriate. These should take into account the required survival and operating modes of systems following a major incident. (For additional loss prevention features of the control or equipment room containing SIS hardware, see Data Sheet 5-32, *Electronic Data Processing Systems*.)

2.3.3 Power Requirements

1. When evaluating the reliability of SIS, consider the security of electrical supplies under plant upset conditions and the effects of partial and complete failure of the main electrical systems. A properly designed uninterruptible power supply or alternate power source, such as an auxiliary generator, may be needed to achieve satisfactory system reliability.

2. Size and rate electrical supplies based on the worst case load with all elements energized including motor start currents and other impulse loads.

3. Provide adequate power supply cooling and ventilation under both normal and upset conditions.

2.3.4 Field Devices

The design, selection and location of sensors and actuators can make a very significant contribution to the overall performance of an SIS. Failure of field devices is a major contributor to overall system unreliability.

1. Mark components of the SIS located in the plant with unique, permanent labels on or near the equipment in accordance with drawings and documentation.

2. Many common cause failures of redundant field devices can be avoided by properly applied diversity of devices. When needed to achieve the desired SIL, obtain diversity by measuring different variables, where each is indicative of the same abnormal condition. For example, a critical tank liquid level could be measured by both a float and differential pressure device. Another option would be using different measurement technologies for the same variable.

3. Analog input devices (e.g., temperature indicating controller) are preferable to switched input devices (e.g., high temperature limit switch). The ability to continuously compare signals reduces the mean time to detection of failure and hence increases integrity.

4. Smart transmitters (I/O devices that are capable of local computations or logic functions) should not negatively impact the SIS. Because they can be easily reprogrammed, strictly control access and protect the SIS by electronic integrity, e.g., read-only communications.

5. Evaluate all shutdown and blowdown valves that are part of the SIS for proper fail-safe mode.

6. Air should be the power medium for shutdown and blowdown valves. However hydraulic or electric fail-safe actuators may be justified for some applications despite their greater system complexity. In all cases, adequate safety integrity and survivability of the valve and associated controls should be ensured.

7. The shutdown and blowdown valves should be capable of being operated under maximum line differential pressure.

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8. Exercise valves, actuators and ancillary devices regularly to ensure reliable operation. For most mechanical devices, this should be at least monthly.

9. Do not provide shutdown valves with hand wheels or positioners. This is to prevent the valve from being in an intermediate or ‘‘control’’ position. They should operate fully open or closed. If hand wheels or positioners are used, the valve must also be provided with limit switches to indicate ***both*** full open and closed position. At any intermediate position, they should alarm in the control room and prevent operation of the system until the valve is returned to its proper position.

10. Where it is necessary to use control valves in a safety related application, e.g., for controlled blowdown of plant to flare, the control valves and their associated systems and ancillary devices should be suitable for the required integrity (SIL) of the application.

2.3.5 Operator Interface

1. The operator interface should be designed using human factors principles. The presentation of information to the operator should be clear and unambiguous. The volume of alarms and messages which will be presented to the operator in a plant upset situation should be assessed and managed.

2. The operator should readily be able to determine the cause of any disturbance or unusual event.

3. The number of control room operators should be determined based on the ability to handle both normal and upset conditions.

2.4 Software

Software is the intellectual creation comprising the programs, procedures, rules and associated documentation pertaining to the operation of a programmable electronic system. It includes system software, firmware (special purpose memory units containing software embedded in protected memory required for the operation of programmable electronics), and application programs. System software includes the PES operating system, telecommunication handling software and device drivers if these software items are part of the standard PES product and are not part of the application software.

Software errors occur because of design faults or omissions or because the function/outcome is improperly specified. Physical failures occur when hardware wears out or breaks. Design faults occur in non-programmable equipment, but the inherent complexity of software systems significantly increases the chance for design faults. The techniques available for achieving high levels of software quality assurance need well defined, change free specifications. These factors should be carefully considered prior to deciding on the use of programmable electronic systems for high SIL applications.

2.4.1 General

1. Software based systems should incorporate an internal log showing the software version or revision giving date and time of the last change and the person who made the change.

2. Regularly make backup copies of the operating system, applications programs and important data and store them in a safe location, not exposed to a fire, explosion or other incident that might affect the operating system.

3. Develop system software according to quality assurance standards which satisfy the safety integrity level requirements of the final application e.g., ANSI/IEEE 730 — *Standard for Software Quality Assurance Plans*. Incorporate evidence of the quality assurance applied to the development of the system software into the software quality plan. This evidence could include reports of design reviews, code reviews, software and integration testing, etc.

4. Develop application software (software within a PES that is specific to a customer application) according to quality assurance standards which satisfy the integrity level of the final application. Ensure the software is readily understood by all technical personnel. It should be readily tested, and be modified in a controlled manner. This means the software should be of minimum size and complexity.

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2.4.2 Data Communication

1. Communication links between the I/O devices and the PES should be hardwired rather than telemetry (wireless). Telemetry is subject to many interference sources (radio frequency, electromagnetic, lightning, etc.) and other limitations that make it unsuitable for critical safety applications.

2. Where necessary for system reliability, prevent physical damage to critical communication links by providing redundant links with diverse routing. Cables for ‘‘energize to execute’’ circuits should avoid high risk or vulnerable areas (e.g., high EMI, high voltage) where practical. Test redundant links regularly.

3. Where communications links exist between the SIS and the BPCS, take steps to prevent corruption of the SIS programs. Read-only access by the BPCS is one method. (See also 2.3.1, item 2)

2.5 System Turnover (Acceptance) Testing

1. Prior to installation, test the logic system for all SIS in accordance with a previously agreed upon test program. Simulated inputs and outputs may be used in testing at the vendor’s site. It should include a complete verification of the operating manuals and related documentation.

2. Full system testing, including all field elements, should be carried out during commissioning. This testing should duplicate the Factory Acceptance Test, with the substitution of actual plant sensors and final control elements. The input devices should be subject to simulated process variables to verify the integrity of the entire control loop. Turnover documentation should include system configuration, logic or functional control diagrams, setpoints, system response times, processor utilization, setpoints and calibration records to establish a baseline for future configuration management and change control.

2.6 System Maintenance and Testing

The purpose of systems maintenance and testing is to ensure that the performance standards set in the initial design are maintained throughout the life of the SIS.

2.6.1 Responsible Person

1. Every SIS should be under the control of an identified responsible person or job position.

2. The responsible person or job position is accountable for ensuring that the systems continue to meet the required performance standards. Specific responsibilities include:

a. Assurance of the competency of the operators and maintenance technicians who work with or on the system.

b. Control of access to the system including use of keys and passwords.

c. Control and documentation of overrides.

d. Coordination of systems testing.

e. Control and documentation of changes to the system.

f. Release of the system for return to service following modification, maintenance and/or testing.

g. Maintenance of appropriate records.

h. Assessment of the results of testing, maintenance activities, system failures, and the demand rate on the system to ensure system integrity is maintained. All deviations are corrected.

2.6.2 Management of Change

1. Have management of change processes in place to control, document and monitor proposed and actual changes to hardware, software, training and operational procedures.

2. Make sure all changes meet the system’s safety performance standard. All relevant parties should fully assess and agree to the changes before implementing them.

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3. Fully test changes to software-based safety systems prior to implementation on an operational host system. Once installed on an operational system, the software should be capable of immediate change back to a known working version in the event an error is detected. The cause of such a reversion should be identified and corrected.

4. Changes impacting the implementation of safety functions will require recertifying the SIS.

2.6.3 Maintenance and Testing

1. The plant operations manual should clearly document design assumptions on the scope and frequency of testing.

2. The plant operations manual should describe how demands on the systems will be recorded and how the systems will be assessed periodically to ensure that their safety integrity meets or exceeds the design performance standards. Any shortfalls will need to be addressed. The implications of any failures should be assessed, and where required, modifications to equipment or procedures should be carried out to minimize the likelihood of repeat occurrences.

3. For large complex systems, consider having a vendor support contract for corrective and preventative maintenance, spares management, and support for system changes.

4. Have the necessary tools and diagnostic facilities available to permit technicians to perform first line maintenance and restore system availability within a reasonable period of time.

5. Provide facilities to enable complete on-line testing of all system components including power supplies and field equipment unless adequate safety integrity can be achieved by testing during planned shutdowns. The objective is to detect and rectify any failures not detected by on-line diagnostics and checking.

6. Fully authorize and record use of maintenance overrides in a manner similar to hot work permits. Records of control system overrides could include date and time bypassed and cleared, reason for bypass, person authorizing the bypass, etc. Use of overrides should be subject to instructions and procedures described in the plant operations manual. The status of overrides should be continuously annunciated at a suitable, continuously occupied operator interface and regularly assessed.

7. Design and control maintenance override facilities, if required, to prevent loss of automatic protection for SIL 3 and 4 systems and minimize protection loss for SIL 1 and 2 systems.

8. Do not permit master bypass of shutdown initiators.

2.6.4 Documentation and Records

1. Make complete and up-to-date documentation available to maintain the system throughout its life cycle. This includes overall system description, performance specifications, key drawings, software and hardware modifications, and training, operation and maintenance instructions and other information found useful by maintainers. Duplicate this documentation at a remote site to recover from a disaster.

2. Keep records of the following throughout the life of the system or for predefined periods as appropriate:

- Inspection records
- Training records
- Testing records
- Maintenance repairs
- System failures
- System modifications & impact analysis of those modifications
- System demands and outcomes
- System integrity assessments
- Subsequent changes to the scope or frequency of testing

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2.6.5 Assessment of SIS Integrity

1. Assess the results of periodic system testing and take appropriate measures to maintain the required system integrity.

2. For SIL 3 and 4 systems, periodic reviews are necessary to ensure that the safety integrity is maintained during the life of the plant. These reviews should re-examine the quantified analyses originally carried out during the design phase taking into account actual demands on the systems, outcomes of those demands, system failure rates, any revised testing regimes, and any changed operational circumstances.

2.7 Control Houses and Equipment Rooms

Control houses (separate buildings containing process control systems and operator panels) and equipment rooms (rooms within a building containing process control electronics and possibly operator panels) can be a critical element in plant operations. These areas should be afforded maximum protection through proper spacing from exposures, explosion resistant construction, and adequate barricading when necessary. Construction should minimize the combustible load present and fire protection systems should be provided where needed.

2.7.1 Location

1. Control house and equipment rooms should be located a safe distance from hazardous processes. (See also Data Sheet 7-44). The determination of a safe distance should be based on a study of the individual characteristics of the equipment and processes that expose the control house/room. These include:

a. Types of materials present in the equipment and types of reactions or processes.

b. Operating pressures and temperatures.

c. Quantities of material present in the equipment.

d. Type of construction of the control house.

e. Value of the control house and importance to production.

f. Fire protection facilities provided for the exposure including reliability in the event of explosion.

g. Drainage facilities, including dikes and slope of ground.

2.7.2 Construction and Protection

1. Control or equipment rooms should be in cut-off noncombustible buildings or areas having no other occupancy, whenever possible. For facilities located in process buildings:

a. The cutoff should be commensurate with the exposure but at least 1-hour fire resistance with self-closing access doors unless the process area is noncombustible with no fire or explosion hazard.

b. If the process area has a room explosion hazard, the cut-off construction should be pressure resistant. (See Data Sheet 1-44.)

2. Where there is unfavorable exposure from outside the building, blank walls of suitable fire, explosion, or missile resistance should be provided.

3. A separate cut-off room of 1-hour fire resistance should be provided for necessary storage of records and operating supplies.

4. Raised floors and suspended ceilings should be constructed essentially of noncombustible materials. Openings into the space under raised floors should be covered or screened to keep out combustible debris. If the spaces created by raised floors or suspended ceilings are used for cables, wires, plastic tubing, or air handling, easily removable panels or other ready means of access should be provided. (See also Data Sheet 5-32 for other protection and prevention recommendations for spaces with cables)

5. If the control house or equipment room is located near a potential source of substantial release of flammable gases or vapors such as LP gas or ethylene, the control area should be maintained under a positive

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air pressure of at least 0.1 in. of water (25 Pa). Fresh air should be taken from a safe location. Under most conditions, this should be at least 30 ft (9 m) above ground.

6. Automatic sprinklers should be installed:

   a. Throughout the control or equipment room, unless construction is entirely noncombustible and there are no cards, paper, or other combustibles in the room except instrument charts and logs in use.

   b. In record storage, supply storage and maintenance areas where the amount of combustibles would warrant such protection.

7. Where the control or equipment room has extensive electronic control or data processing equipment, refer to Data Sheet 5-32 for additional loss prevention advice.

8. Portable 10 to 15-lb. (4-6 kg) clean agent or carbon dioxide fire extinguishers should be provided for fires in electrical equipment. Avoid the use of dry chemical extinguishers. Provide 2-1/2 gal. (10 liter) water-type fire extinguishers for fires in ordinary combustibles in the control room or associated storage rooms.

3.0 SUPPORT FOR RECOMMENDATIONS

3.1 Comments and Technical Support

3.1.1 Overview of Instrumentation and Control

A typical basic process control system (BPCS) and safety instrumented (control) system (SIS) is shown in Figure 4. It is important to note that there are ***two independent functions***, the control of the process (BPCS) and its safety (SIS). Process control systems ensure the quality and quantity of the product and frequently reduce the need/reliance on an operator to produce the product. The safety system functions when the BPCS fails to perform. A well designed control system can significantly reduce the demands on the safety system and reduce the overall risk presented by the process.

Control and safety systems play a significant role in the management of hazards in the process industry. Safety systems are recognized for reducing the likelihood and consequences of events involving potential damage of assets and production interruption. The role of these systems could include prevention of overpressure, limiting high and low levels, detection of fire and flammable or toxic gas release, shutdown and isolation of plant and inventories, and initiation of firefighting systems.

It is increasingly recognized that design safety should be integrated across engineering disciplines and throughout the conceptual and detailed design process. Identifying and eliminating hazards should be an early objective in the design process.

Figure 5 is a flow chart showing this review and design process and its relationship to process safety systems. A similar safety life cycle model appears in ANSI/ISA S84.01. This data sheet addresses issues related to the safety systems portion of the flow chart. Other steps in the safety life cycle may be addressed by other FMRC documents as well as by outside standards.

Factory Mutual will accept widely recognized standards of the Instrument Society of America, (ANSI/ISA S84.01 - *Application of Safety Instrumented Systems for the Process Industries*) and the International Electrotechnical Committee (IEC Draft Publication 61508 - *Functional Safety: Safety Related Systems*) as minimum requirements for SIS when used in chemical plants or other facilities that are to be considered Highly Protected Risks (HPR). The ANSI/ISA standard is somewhat industry specific (process industries) while the IEC standard attempts to address installation of all safety related electronic systems in any industry. (The IEC has application specific standards under development.)

The ISA Standard defines SIL and other mandatory requirements, however, it does not prescribe a particular SIL to be provided for individual processes. This is left to the judgment of the user and installer of the equipment. This philosophy runs parallel to that of Process Safety Management (OSHA 29 CFR 1910.110 or similar national or international standards) which has performance-based requirements as opposed to specification-based requirements. This data sheet provides a method to be used in selecting the SIL needed for a particular process or operation. This methodology depends on the risk (function of consequence and likelihood of an excursion) associated with the process under study.

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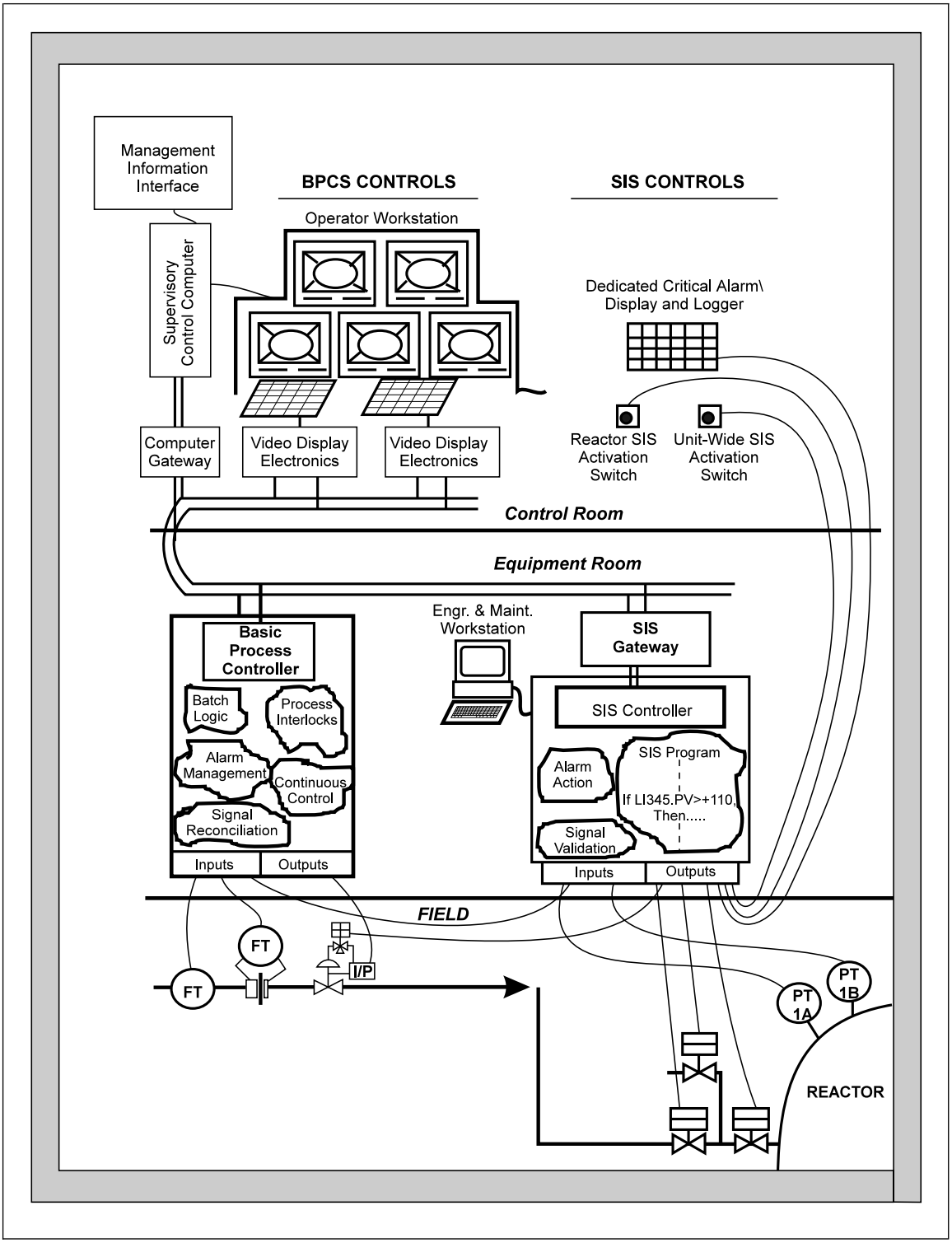


Fig. 4. Typical control system including BPCS and SIS components.[1]

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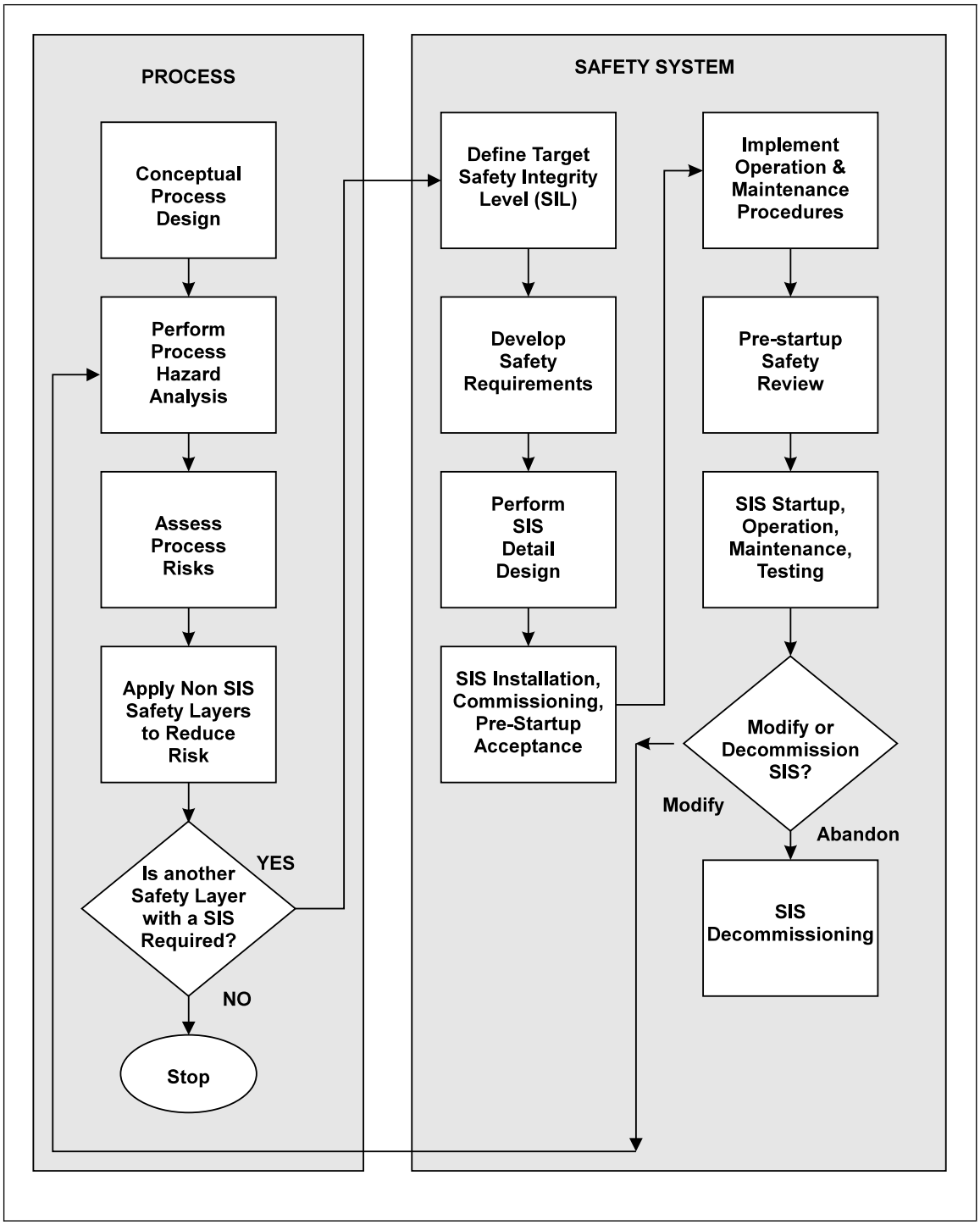


Fig. 5. Safety lifecycle model.

In the past, chemical processing facilities used analog circuitry for the instrumentation and control of chemical operations and processes. Due to advances in computer technology, control and safety systems are being re-evaluated and newer Programmable Electronic Systems (PES) are being installed. These systems are capable of performing safety functions quickly, precisely and with a high degree of consistency.

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There are drawbacks to systems of this type in that one failure in the system may impact many different parts of the overall system. There is a common perception that provision of a PES alone will provide higher levels of reliability and safety, however, this is only true when the system has been designed taking into consideration the reliability of the individual elements of the PES. The reliability of the input devices, logic processors and output devices must be studied as a system to arrive at an accurate evaluation of the system's reliability and safety.

3.1.2 Control System Versus Safety System — Protection Layers

Chemical processes are provided with several layers of protection (Fig. 2) to prevent an excursion or reduce its consequences.

The first layer involves the basic process design where steps can be taken to minimize the exposure to an incident. The ideal is an inherently safer system. (An inherently safer design ***avoids*** hazards by removing or reducing the amount of hazardous material in the plant or the number of hazardous operations).

The next layer, commonly called the basic process control system (BPCS), includes the operator and the process control system. This layer is predominantly a quality assurance measure (e.g., smooth running operations result in higher quality and quantity of product). The operator monitors and adjusts process variables such as temperature, pressure, flow rate, etc. to meet process needs. The control system is expected to maintain the process at the set points established by the operator and within safe limits of operation.

Another layer involves critical alarms and operator supervision. This could be an extension of the control system and at this level, manual intervention of the operator contains the excursion.

The next layer is the safety system that responds if and when the previous layer fails. This layer is often called a Safety Instrumented System (SIS) or Emergency Shutdown System (ESD).

The next layer is passive physical protection provided for the process (e.g., pressure relief devices). Designed to minimize the extent of damage during a major upset, this layer is the final protection layer prior to an out-of-control situation. Once the event has occurred, other layers such as containment dikes, fire walls, emergency drainage and an on-site or public fire brigade are provided to minimize the impact on the business and community.

The concept for evaluating these safety layers is independent protection layer (IPL). A protection layer qualifies as an IPL when:

1. The protection reduces the identified risk by a large amount, that is, at least a 100-fold reduction.
2. The protection also has the following characteristics:

   a. *Specificity*: The IPL is designed solely to prevent or mitigate the consequences of one hazardous event (e.g., a runaway reaction, release of hazardous material, loss of containment or a fire). Multiple causes may lead to the same hazardous event and therefore multiple event scenarios may initiate action of one IPL.

   b. *Independence*: The IPL is independent of other protection layers associated with the identified danger.

   c. *Dependability*: The IPL can be counted on to do what it was designed to do. Random and systematic failure modes were addressed in the design.

   d. *Auditability*: The IPL is designed to facilitate regular validation of its protective function by maintenance or functional testing.

The BPCS functions as a protective layer but this multipurpose instrument system fails the test of *specificity* because its first purpose is to regulate day to day operation. Modifications are routinely made, often without follow-up validation of functional integrity.

Items that typically receive credit as IPLs include pressure relief devices (rupture disk, relief valves) and shutdown valves. Sprinklers or protection systems involving an alarm followed by operator response are not considered to have the 100-fold ***likelihood*** reduction and hence are not considered IPLs.

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For example, a fire box explosion on a boiler is prevented by the fuel gas train which consists of a number of interlocks (high and low gas pressure switches, low combustion air pressure switch, etc). The interlocks open the main gas valve through the combustion safeguard. This boiler has one IPL (the entire combustion safety system) to prevent the fire box explosion.

In evaluating the Risk of a proposed or existing process, the analysis must account for those levels present or proposed and come up with the overall process Risk. If it meets predefined acceptable levels, the design is complete. If not, additional or improved layers are needed and the new arrangement is re-analyzed until acceptable Risk is achieved.

3.1.3 Risk

As used in this document, Risk is a product of the Consequence [C] and the Likelihood (P_a) of an accident occurring.

$$\text{Risk} = C \times P_a$$

For FM purposes, Consequence [C] is defined as the potential dollar value loss associated with the damage caused as a result of the process excursion. This depends on the expected severity of the event under study. Included would be the property damage (PD) and the operational shutdown (TE) associated with the event. For FM purposes, costs of harm to persons is not included but it may be a consideration required by other authorities. For example, an increase in pressure within a vessel which simply operates a rupture disc will require a short period of downtime to replace the disc. However, overpressure and catastrophic rupture of a vessel resulting from improper design or failure of the rupture disc will extensively damage adjacent equipment and construction. This will require extensive time for demolition, engineering, reconstruction and re-commissioning.

The magnitude of the Consequence has been divided into three categories of increasing severity as indicated in Table 1.

Table 1. Consequence—Typical Categories

Category	*Description*
Major	Substantial or total destruction beyond a local process area into the surrounding plant and shut down more than 3 months
Severe	Substantial damage mainly to a single process area and shutdown from 2 weeks to 3 months
Minor	Equipment damage and shutdown less than 2 weeks

The likelihood (P_a) of an accident is defined as the product of probability of having a process upset condition (P_{proc}) and simultaneously the probability of a PES failing to respond to the upset condition (P_{pes}).

$$P_a = P_{proc} \times P_{pes}$$

Process upset and PES failure on demand are independent events so, statistically, the resulting likelihood is multiplicative.

For instance, a process may have a known probability of upset of 5×10^{-4}/year (P_{proc}) and the probability of the PES failure on demand (P_{pes}) is determined to be 1×10^{-3}/demand. Therefore, the likelihood (P_a) is calculated as 5×10^{-7}/year [$(5 \times 10^{-4}) \times (1 \times 10^{-3})$].

Values for P_{proc} are typically determined by conducting a Quantitative Risk Assessment (QRA) for the process or may be part of documentation from many years of data gathering and evaluation. The values for P_{proc} and P_{pes} are frequently unavailable or extremely cumbersome and time consuming to produce. Several groups are currently collecting failure probability data. For the purposes of this data sheet, Table 2 (a generalization; *qualitative* analysis rather than *quantitative*) should be used in the absence of quantitative data.

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Table 2. Probability of Process Upsets—Typical Categories

Probability	*Type of Initiating Event*
High	Failure can be expected within the life of the plant. *Examples include process leaks, single instrument or valve failures, or human errors which result in releases of hazardous material*
Moderate	Failure or series of failures may occur with low probability within the life of the plant. *Examples include dual instrument or valve failures, a combination of instrument failure and human error or large releases in loading/unloading areas.*
Low*	Failure may occur with very low probability within the life of the plant. *Examples include combinations of multiple instrument failures and multiple human errors, or full-bore failures of small process lines or fittings.*

* Multiple instrument or valve failure, multiple human errors, or spontaneous failures of vessels have lower likelihood but may also be considered in this group.

3.1.4 Safety Integrity Level (SIL)

Safety systems are classified based on the probability of failure on demand, P_{pes}. This means the system will fail to respond appropriately when the demand occurs. The lower the probability of failure on demand, the higher the availability of the system to act in the manner for which it was designed.

Availability = 1 - Probability of Failure on Demand (P_{pes})

SIL is a function of the configuration and redundancy of not only the logic processor, but also the input and output devices, including all field devices (Fig. 1). Having two logic processors instead of one will decrease the probability of failure on demand of this component. Therefore, the availability of the system has been increased by some factor. As redundancy is provided throughout the whole system (e.g., sensors, logic processors, output devices, i.e., Fig. 6), the probability of failure on demand is reduced in the short term and therefore it will have a higher availability to respond in an appropriate manner. This results in a higher SIL. A system having redundant logic processors but single input or output devices will have a lower SIL than a system provided with redundancy for all elements of the system.

(Note: As the system complexity is increased, there are more components to fail so the long term availability may decrease. Such trade offs have to be evaluated based on the needs of the individual system.)

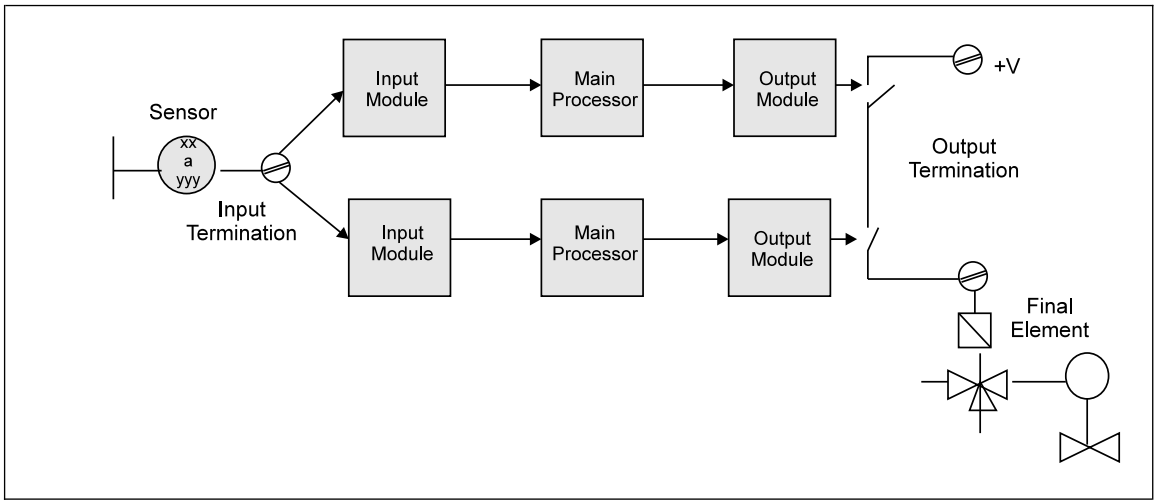


Fig. 6. Example of possible SIL 2 architecture.

FMRC has adopted the IEC 61508 definition of SIL as indicated in Table 3. ANSI/ISA S84.01 uses a similar classification but excludes SIL 4. SIL 4 systems can be found in avionics and nuclear applications but are not typical in the chemical process industries.

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Table 3. Classification of Programmable Electronic Systems Based on Probability of Failure

SIL	*Probability of Failure on Demand*	*Safety Availability*
4	10^{-5} — 10^{-4}	99.999 — 99.99
3	10^{-4} — 10^{-3}	99.99 — 99.9
2	10^{-3} — 10^{-2}	99.9 — 99
1	10^{-2} — 10^{-1}	99 — 90

3.1.5 How is System SIL Determined?

Using reliability analysis techniques, the overall availability of a safety system can be mathematically computed and a SIL assigned. This method uses all possible states (combinations) of the safety system from the fully operational, to partially failed to completely failed. Analysis is performed via complex computer applications and a computer program has been developed by FMRC to help evaluate any safety system architecture.

For the evaluation of safety systems, the quantitative SIL value should be used. If it is not available, the generalizations shown in Table 4 can be used for preliminary screening purposes. Many factors can affect the reliability (SIL) of a particular system include redundancy, diversity, voting, diagnostics and maintenance and testing frequency and any ‘‘adequacy for service’’ decisions should be based on a detailed evaluation of an existing or proposed system.

Table 4. SIS Interlock Design Structure for Each Integrity Level

Integrity Level	*Minimum Design Structure*
1	*Non-redundant*. Best single path designs, e.g., Figure 1.
2	*Partially redundant*. Redundant independent paths for elements of lower availability. There may be some diagnostic capabilities, e.g., Figure 6.
3	*Totally redundant*. Redundant independent paths for the total interlock system. Diversity, separation, redundancy and exhaustive diagnostic capabilities are key aspects of this level. A single fault in an SIS component is unlikely to result in loss of process protection, e.g., Figure 7.

Note: A nonprogrammable, relay based safety system, common in smaller boilers for combustion safety, could be considered equivalent to a SIL 1 programmable system.

3.1.6 SIL Versus Inspection Frequency

Using reliability analysis techniques, it is possible to produce a graph which denotes the availability of the system as a function of inspection frequency. Figure 8 shows a typical representation. As would be expected, performance of the safety system deteriorates over time without periodic inspection and testing. These graphs should be available from the system vendor upon completion of certification. This graph is used to determine what inspection interval is needed to achieve the desired SIL.

As can be seen from Figure 8, System 5 would require extremely frequent inspections to achieve SIL 4. This inspection frequency is most likely unrealistic in terms of real world operations. System 1, however, would allow a significant time between inspections to meet SIL 4.

3.1.7 Determining SIL Required for an Application

Once the Consequence [C] and the probability of a process upset (P_{proc}) are determined, the level of safety control needed (SIL) can be determined from the SIL Matrix (Fig. 3).

1. Determine the Consequence of an Incident

The Loss Expectancy (PD & TE) of an out-of-control excursion of the operation or process is the first step in determining the required SIL. If there are multiple operations or processes utilizing individual safety systems, an analysis will be required for each process and safety system combination. If several operations

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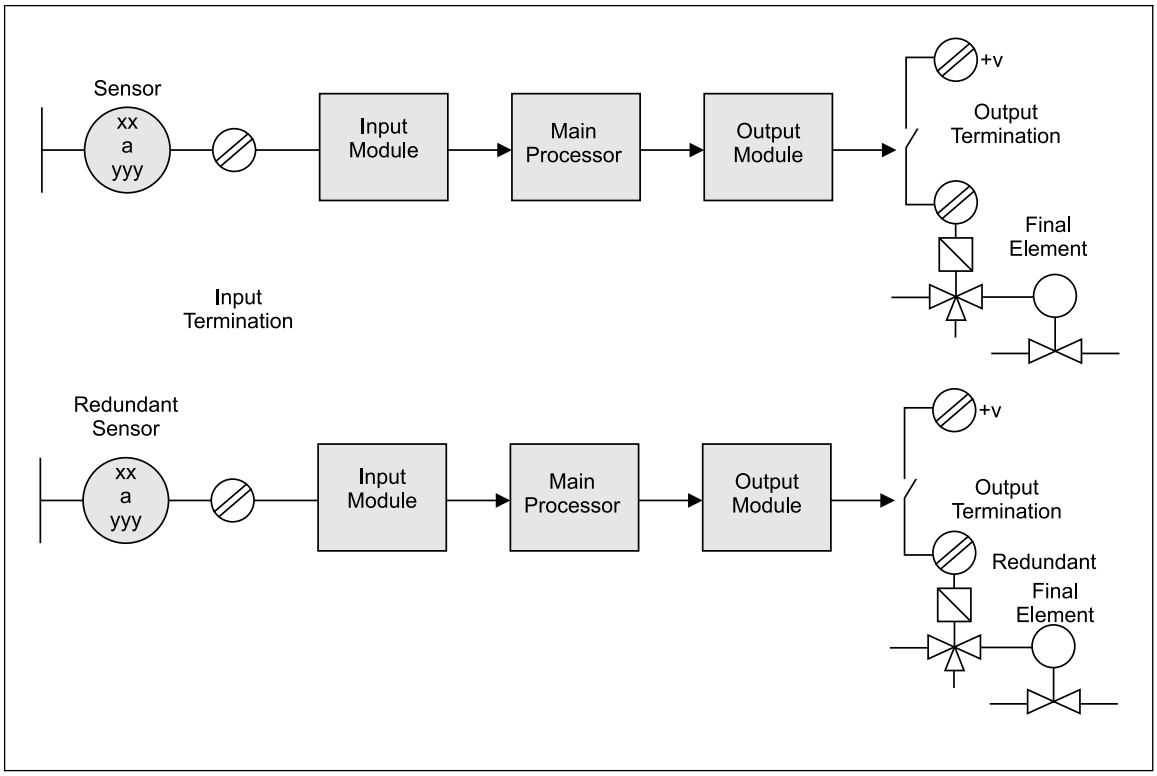


Fig. 7. Example of possible SIL 3 architecture.

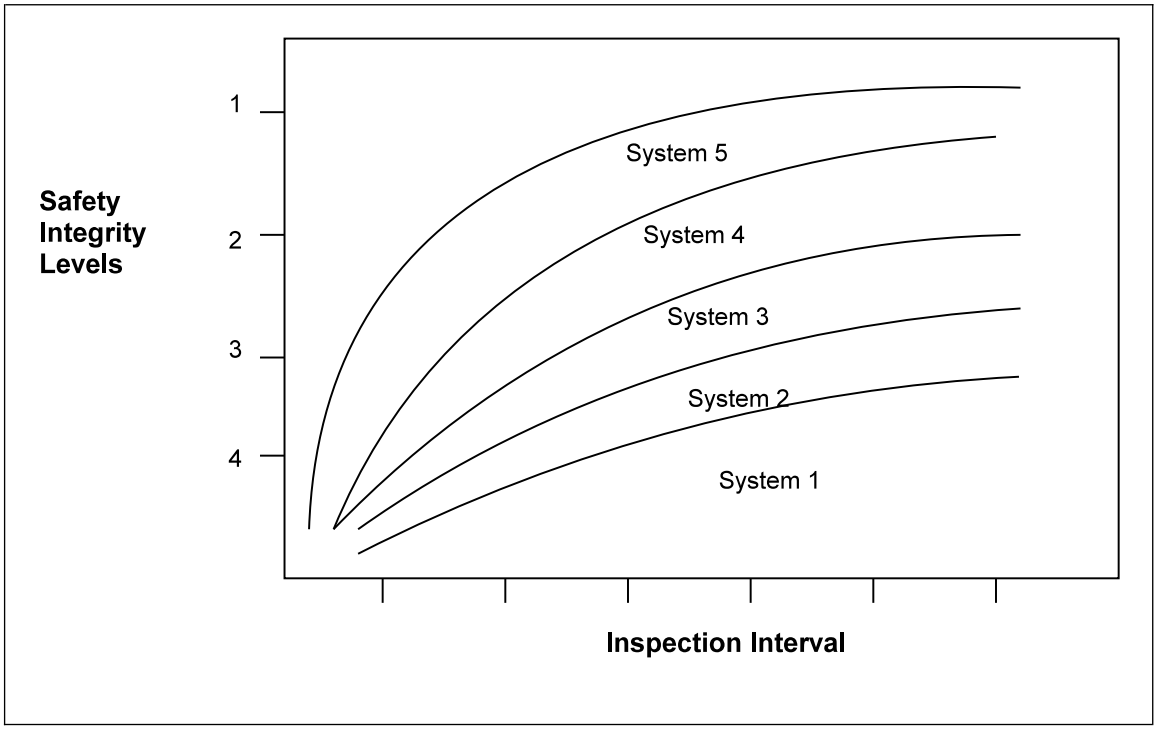


Fig. 8. Effect of inspection interval on SIL.

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or processes are under the control of a single safety system, an analysis of each operation or process (individually or in combination) will be needed to determine which scenario results in the highest required SIL. This will control the design of the entire safety system.

The Consequences will depend on the degree of hazard and the expected results from a process excursion. The following components should be evaluated:

a. Expected Property Damage including:

- Building Damage
- Machinery & Equipment Damage
- Stock & Supplies

b. Costs associated with demolition, cleanup and possible government intervention for investigation.

c. Costs associated with re-design

d. Lost production as a result of the incident and ensuing period required for rebuild. The amount of time required for demolition, cleanup, redesign, construction and commissioning the operation should be considered.

2. Determine the Incident Probability

To evaluate incident probability, a Quantitative Risk Assessment (QRA) should be performed on the operation/process being evaluated. This analysis might be available as a result of facility's compliance with national or local regulatory requirements. If this variable is not known or available, a qualitative evaluation can be made. The scenario under study will need to be evaluated to determine which failures will result in a process upset. The number and types of failures necessary can then be used to determine a qualitative probability of process upset from Table 2.

3. Determine the SIL needed

The SIL Matrix (Fig. 3) can be used to determine the need for an SIS to protect the risk from a specific process upset by following the following steps:

a. Categorize the consequence of the process upset using Table 1.

b. Identify the probability of a process upset using the categories in Table 2.

c. Identify the number of non SIS IPLs against the process upset already existing or planned.

d. With the information from steps 1-3, enter the SIL Matrix (Fig. 3) first picking the consequence level and within that level, the likelihood. At that point, go up to the number of other IPLs and determine if an SIS is needed to reduce the risk from a particular process upset and if needed, what SIL is required.

4. Example

A fire box explosion on a boiler is prevented by the fuel gas train which consists of a number of interlocks (high and low gas pressure switches, low combustion air pressure switch, etc). The interlocks open the main gas valve through the combustion safeguard. This boiler has one IPL (the entire combustion safety system) to prevent the fire box explosion.

An evaluation of the effect of a boiler explosion concludes that consequences would be rated Severe. The probability of an incident would rate Low. In Figure 3, no additional SIS is required. (Enter Figure 3 with Consequence = Severe, Probability = Low, Number of non SIS IPLs = 1 and the Matrix block is blank)

3.2 Certification of Safety Systems

Factory Mutual Research Corporation (FMRC) can provide third-party certification for a safety system and assign a corresponding SIL for the system as a whole. Vendors can submit products (SIS components) to FMRC for evaluation with subsequent listing in the *Factory Mutual Approval Guide* as fit for use in the implementation of safety functions of a given SIL. This service is also available for installed equipment on a case-by-case basis.

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3.3 "Soft PLC" Systems

There is information that current PLC systems, as addressed in this data sheet, will have competition from "soft PLCs". These are being promoted by several vendors and the subject of articles in technical magazines. Essentially, this will be a control system run by off-the-shelf personal computers. In this application, the system would need to be running at least on a Windows NT® operating system or one specifically designed for control applications. Lower level operating systems (Windows® 98, 95, etc) are not stable/reliable enough for a control environment.

Such systems would have to comply with all of the requirements specified by this data sheet and probably would not be justified for a safety application. Problems would include equipment reliability, access control, control and documentation of changes, software errors and quality assurance of control programs, etc. Any such application should be evaluated critically and for the most part would only be used in control applications that present insignificant loss exposure, for example, control of individual research reactors, single production machines in metal working or food processing, etc.

4.0 REFERENCES

1. Center for Chemical Process Safety (CCPS), *Guidelines for Safe Automation of Chemical Processes*, New York, 1993.

2. Instrument Society of America, ANSI/ISA S84.01 — *Application of Safety Instrumented Systems for the Process Industries*

3. International Electrotechnical Committee, Draft Publication 61508 - *Functional Safety: Safety Related Systems*

4. ANSI/IEEE 730 — *Standard for Software Quality Assurance Plans*

5. ANSI/IEEE 828 — *Standard for Software Configuration Management Plans*

6. ANSI/IEEE 829 — *Standard for Software Test Documentation*

4.1 Other Comparable National Standards

DIN V VDE 0801 — *Principles for Computers in Safety-related Systems*

DIN V 19250 — *Basic Safety Assessment for I&C Protection Devices*

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